3. Statistics (Sufficiency and Hypothesis Testing)

1. Sufficient statistics
2. What is hypothesis testing
3. Elements of a test
4. Methods of finding tests
    - Neyman-Pearson paradigm
    - Generalized likelihood ratio test
5. Testing mean and variance in Normal distribution

Reading: Rice Ch9, 11.2-11.3; Casella and Berger Ch8, 10.3.

$g(x_1, \cdots x_n) \Rightarrow \theta$

Information and Data Reduction

- (Numerical or graphical) transformations of data appear all the time in statistics for offering a summary of information contained in data. For example
  - Raw (original) data $X_1, \cdots, X_n$
  - Order statistics $X_{(1)}, \cdots, X_{(n)}$
  - Histogram
  - Sample mean $\bar{X}$
  - Sample variance $\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$
- Data reduction through transformation is required to present concrete information. However, it can cause the loss of information.

- Question: Is there a statistic $T(X_1, \cdots, X_n)$ which contains all the information in the sample about parameter θ ?

$f(x_1,\cdots,x_n \mid T=t)$ free of θ.

Sufficient Statistics

$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$ $\quad$ $\frac{X_1-\mu}{\sigma} \sim N(0,1)$

- Sufficient statistics: A statistic $T(X_1,\cdots,X_n)$ is said to be sufficient for parameter θ if the conditional distribution of $X_1,\cdots,X_n$ given $T=t$ does not depend on θ for any value of t.

MLE. $\hat{\mu}=\bar{X}$

- Example 4.1. $X_1,\cdots,X_n \sim N(\mu,\sigma_0^2)$ i.i.d., where σ_0^2 is known. Then the sample mean $\bar{X}$ is a sufficient statistic for μ.

$\bar{X} \sim N(\mu, \frac{\sigma_0^2}{n})$

Soln: Consider conditional distribution of $X_1,\cdots,X_n$ given $\bar{X}$

$f_{X_i}(x|\mu)=\frac{1}{\sqrt{2\pi\sigma_0^2}}e^{-\frac{(x-\mu)^2}{2\sigma_0^2}}$

$P(X_1\le x_1,\cdots,X_n\le x_n, \bar{X}\le\bar{x})$

$$f(x_1,\cdots,x_n|\bar{X}=\bar{x})=\frac{f(x_1,\cdots,x_n)}{f_{\bar{X}}(\bar{x})}.$$

$=P(X_1\le x_1,\cdots,X_n\le x_n)$

$f_{\bar{X}}(\bar{x})=\frac{1}{\sqrt{2\pi\sigma_0^2/n}}e^{-\frac{n(\bar{x}-\mu)^2}{2\sigma_0^2}}$

$$=\frac{\prod_{i=1}^n f(x_i|\mu)}{f_{\bar{X}}(\bar{x}|\mu)}=\frac{(2\pi\sigma_0^2)^{-\frac{n}{2}}e^{-\frac{\sum(x_i-\mu)^2}{2\sigma_0^2}}}{(2\pi\sigma_0^2)^{-\frac{1}{2}}\cdot n^{\frac{1}{2}}\cdot e^{-\frac{n(\bar{x}-\mu)^2}{2\sigma_0^2}}}$$

$$=(2\pi\sigma_0^2)^{-\frac{n-1}{2}}n^{-\frac{1}{2}}\cdot e^{-\frac{1}{2\sigma_0^2}\cdot\sum(x_i-\mu)^2-n(\bar{x}-\mu)^2}$$

$\Downarrow$

$\sum x_i^2-n\bar{x}^2$.

free of μ.

$\sum(x_i-\mu)^2-n(\bar{x}-\mu)^2$
$=\sum x_i^2-2\cancel{\sum x_i\mu}+\cancel{n\mu^2}$
$-n\bar{x}^2+\cancel{2n\bar{x}\mu}-\cancel{n\mu^2}$

- **Example 4.2.** Let $X_1, \cdots, X_n$ be i.i.d. Bernoulli (θ) random variables with $P(X_i = 1) = \theta$. $\theta^{x_i}(1-\theta)^{1-x_i}$
  Let $T = \sum_{i=1}^{n} X_i$. Then $\prod_{i=1}^{n} P(X_i = x_i)$ $t = \sum x_i$

joint pmf

$$
\begin{aligned}
P(X_1 = x_1, \cdots, X_n = x_n | T = t) &= \frac{P(X_1 = x_1, \cdots, X_n = x_n, T = t)}{P(T = t)} \quad \sum x_i \\
&= \frac{\theta^t (1-\theta)^{(n-t)}}{\binom{n}{t} \theta^t (1-\theta)^{(n-t)}} = \frac{1}{\binom{n}{t}},
\end{aligned}
$$

free of θ.

if $x_1 + \cdots + x_n = t$ and x_i are nonnegative integers, and 0 otherwise. The conditional distribution is free of θ. So T is sufficient for θ.

① $X_1 \cdots X_n \sim Exp(\lambda)$ iid. $f(x|\lambda) = \lambda e^{-\lambda x}$.

$f(x_1 \cdots x_n|\lambda) = \prod_{i=1}^n f(x_i|\lambda) = \lambda^n e^{-\lambda \sum x_i} = \underbrace{g(T(\vec{x}), \lambda)}_{\lambda^n e^{-\lambda \sum x_i}} \cdot \underbrace{h(\vec{x})}_{\equiv 1}$

How to find a sufficient statistic

$X_1 \cdots X_n$ iid. pdf $f(x|\theta)$

- Factorization Theorem: A necessary sufficient condition for $T(X_1, \cdots, X_n)$ to be sufficient for a parameter θ is that the joint pdf or pmf factors in the form

$$f(x_1, \cdots, x_n|\theta) = \boxed{g(T(x_1, \cdots, x_n), \theta) h(x_1, \cdots, x_n).}$$

$\hat{\theta} = \hat{\theta}(T(\vec{x}))$.

$\hat{\theta} = \hat{\theta}(T(\vec{x}))$. $\arg\max_\theta L(\theta|\vec{x}) = \arg\max_\theta f(x_1 \cdots x_n|\theta) = \arg\max_\theta g(T(\vec{x}), \theta)$

- Corollary (MLE and sufficient statistic). If T is sufficient for θ, then the maximum likelihood estimate for θ is a function of T.

② $X_1 \cdots X_n \sim Pois(\lambda)$. pmf. $p(x|\lambda) = \frac{\lambda^x}{x!} e^{-\lambda}$

$p(x_1 \cdots x_n|\lambda) = \prod_{i=1}^n \frac{\lambda^{x_i}}{x_i!} e^{-\lambda} = \frac{\lambda^{\sum x_i}}{\prod_{i=1}^n x_i!} e^{-n\lambda} = \underbrace{\lambda^{\sum x_i} \cdot e^{-n\lambda}}_{g(T(\vec{x}), \lambda)} \cdot \underbrace{\left(\prod_{i=1}^n x_i!\right)^{-1}}_{h(\vec{x})}$

$T(\vec{x}) = \sum_{i=1}^n x_i$ sufficient stat for λ.

ΣX_i sufficient for μ.
ΣX_i^2 sufficient for σ^2

$(\Sigma X_i, \Sigma X_i^2)\begin{pmatrix} \frac{1}{n} \\ 0 \end{pmatrix}$

$\Sigma x_i^2 - \left(\frac{1}{n}\Sigma x_i\right)^2 \Rightarrow S^{*2}$

- **Example 4.2. (revisit)** For Bernoulli r.v. $X_1, \cdots, X_n$, we have

$$f(x_1,\cdots,x_n|\theta) = \prod_{i=1}^{n} P(X_i = x_i) = \prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i}$$
$$= \left(\frac{\theta}{1-\theta}\right)^{\sum_{i=1}^n x_i} (1-\theta)^n.$$

Let $t = \sum_{i=1}^n x_i$, $g(t,\theta) = \left(\frac{\theta}{1-\theta}\right)^t (1-\theta)^n$ and $h(x) \equiv 1$. By factorization theorem, $T = \sum_{i=1}^n X_i$ is sufficient for θ.

MLE. $\hat{\mu} = \bar{X}$, $\hat{\sigma}^2 = S^{*2}$

- **Example 4.3 (Sufficient statistic for $N(\mu, \sigma^2)$)** If $X_i \sim N(\mu, \sigma^2)$, $i = 1, 2, \cdots, n$ are i.i.d., where μ, σ^2 are unknown. Then $(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2)$ is a 2-dimensional sufficient statistic for (μ, σ^2).

$\theta = (\mu, \sigma^2)$

$$f(x_1,\cdots x_n|\mu,\sigma^2) = (2\pi\sigma^2)^{-\frac{n}{2}} \cdot e^{-\frac{1}{2\sigma^2}\sum_{i=1}^n (x_i-\mu)^2}$$
$$= (2\pi\sigma^2)^{-\frac{n}{2}} \cdot e^{-\frac{1}{2\sigma^2}[\Sigma x_i^2 - 2\Sigma x_i \mu + n\mu^2]}.$$
$$= \underbrace{(\sigma^2)^{-\frac{n}{2}} e^{-\frac{1}{2\sigma^2}[\Sigma x_i^2 - 2\Sigma x_i \mu + n\mu^2]}}_{g(T,\theta)} \cdot \underbrace{(2\pi)^{-\frac{n}{2}}}_{h(\vec{x})}.$$

$T(\vec{x}) = (\Sigma X_i^2, \Sigma X_i)$

$g(T(\vec{x}), \theta)$

Sample $X_1 \cdots, X_n$ ← dist $F(x|\theta)$

$g(X_1 \cdots X_n) \to \theta$

Introduction to Hypothesis Testing

$H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$.

- Question: What is hypothesis testing?
- Example 4.4. $X_1, \cdots, X_n \sim N(\mu, \sigma^2)$ i.i.d., then

$\bar{X} \sim N(\mu, \sigma^2/n)$

Function of sufficient stat $(\Sigma X_i, \Sigma X_i^2)$

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

μ_0

$Z = \frac{\bar{X} - \mu_0}{\sigma_0/\sqrt{n}} \sim N(0,1)$ under H_0.

Which of hypothesis $H_0 : \mu = \mu_0$ or $H_1 : \mu \neq \mu_0$ is true (μ_0 is given). Under H_0, we have

under H_0

$|T| > 2.23$.

statistic

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}.$$

$|T|$ big violate H_0

$|T|$ small support H_0

If we take e.g., $n = 11$, then $P(|T| < 2.23) = 95\%$ when H_0 is true. (Note $t_{10}(\alpha/2) = 2.23$ for $\alpha = 0.05$) Given a particular sample,

- If the value of $|T|$ is very big (say 5.45), T is not likely to have t_{10} distribution. So H_0 might not be true, reject H_0.
- If the value of $|T|$ is small say $T \in (-2.23, 2.23)$, we have no evidence to reject H_0.

Strategy of hypothesis testing

- Translate the scientific hypothesis into H_0 and H_1
- begin with the assumption that H_0 is true
- Collect data
- Determine whether or not the data contradict H_0

e.g. $\theta = \binom{\mu}{\sigma^2}$ $H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$

$\omega_0 = \{ \binom{\mu_0}{\sigma^2} : \sigma^2 > 0 \}$. $\omega_1 = \{ \binom{\mu}{\sigma^2} : \mu \neq \mu_0, \sigma^2 > 0 \}$.

Elements of a test

- Hypotheses: Statements about population parameters. Let $X \sim f(x|\theta)$, $\theta \in \Omega = \omega_0 \cup \omega_1$, we can consider the following two hypotheses: parameter space.

$$H_0 : \theta \in \omega_0 \text{ versus } H_1 : \theta \in \omega_1.$$

H_0 is referred to as the *null hypothesis* and H_1 is the *alternative hypothesis*. $\omega_0 = \{\theta_0\}$.

- For example, $H_0 : \theta = \theta_0$; $H_0 : \theta \leq \theta_0$; $H_0 : \theta \geq \theta_0$.
Alternatively, $H_1 : \theta \neq \theta_0$; $H_1 : \theta > \theta_0$; $H_0 : \theta < \theta_0$.
Two-sided hypothesis

In hypothesis testing, we determine whether H_0 or H_1 is the best explanation for the data.

- There are two ways we could go wrong:

Table: Decision Table

Decision	True state of Nature: H_0 is true	True state of Nature: H_1 is true (H_0 is false,)
Reject H_0	Type I error (reject truth)	Correction detection
Do not reject H_0. Reject H_1	Correct decision	Type II error (accept false.)

$\alpha = P(\text{Type I error}).$

- We would like to have both Type I error, denoted by α, and Type I error, denoted by β, be a small, but there are tradeoffs.

$\beta = P(\text{Type II error}).$

Best test should have both $\min(\alpha)$. & $\min(\beta)$.

• fixed $\alpha = 5\%$, $\min \beta$.

Other terminology

- We always control Type I error under a certain level, e.g., 0.05, which is called significance level or size of the test.
- We want to find a test to minimize the probability of Type II error, equivalently to maximize the power

$1 - P(\text{do not reject}_{H_0} \mid \overset{H_0 \text{ is false}}{H_1 \text{ is true}}) = P(\text{reject } H_0 \mid H_0 \text{ is false})$

$$\text{power} = 1 - P(\text{Type II Error}) = 1 - \beta$$

- Test statistic: The statistic used to discriminate between H_0 and H_1, say T in Example 4.4. $T \sim$ dist (known) under H_0.
- Null distribution: The probability distribution of the test statistic when the null hypothesis is true, say t_{n-1} distribution in Example 4.4.

$X_1 \, --- \, X_n$ binary. $\sim$ Bern(p).

- **Example 4.5.** Consider a test of $H_0 : p = 0.4$ versus $H_1 : p \neq 0.4$ based on a random sample of size 10 from $Bernoulli(p)$. Let $Y = \sum_{i=1}^{n} X_i$. If the rejection rule is to reject H_0 when $Y \leq 0$ or $Y \geq 8$, then the test size is

$Y \sim$ Bino (n, p)

$\sim$ Bino (n, p_0) under H_0.

P(Type 1 error) = P(reject H_0 | H_0 true) = P($Y \leq 0$ or $Y \geq 8$ | $p = 0.4$)

$$\alpha = 1 - P(1 \leq Y \leq 7 | p = 0.4) = 0.0183.$$

$Y \sim$ Bino $(10, 0.4)$

The Type II error probability and the power are

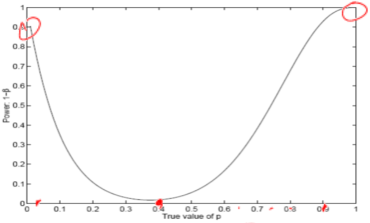
P(Type II error) = P(do not reject H_0 | H_1 is true)

$$\beta = P(1 \leq Y \leq 7 | p \neq 0.4)$$

$p = 0.01, 0.02, ---, 0.49$

and
$$\text{power} = 1 - \beta = 1 - P(1 \leq Y \leq 7 | p \neq 0.4).$$

power.



p.

Neyman-Pearson Paradigm

$\begin{cases} H_0: \underline{\theta = \theta_0} \checkmark \\ H_1: \theta = \theta_1 \checkmark \end{cases}$ dist $f(x|\theta_0)$

$H_0: \underline{X \sim U(0,1)}$

- Simple hypothesis: Completely specify the joint distribution of the random sample. E.g., $H_0 : X \sim Bino(25, 1/3)$.
- Likelihood ratio: Suppose there are two models with parameter(s) θ_0 and θ_1. Likelihood ratio

$L(\theta_0|\vec{x})$

$$\Lambda(\mathbf{x}) = \underbrace{L(\theta_0)}_{H_0}/\underbrace{L(\theta_1)}_{H_1},$$

$\Lambda(\vec{x})$ big if supports H_0, small if violates H_0

where likelihood $L(\theta) = f(\mathbf{x}|\theta)$, $\mathbf{x}$ is the observed data and θ is the parameter(s) in the model.

- Question: What happens if Λ is small?

- Most powerful (MP) test: A test of a simple H_0 versus a simple H_1 is the most powerful test of size α if no other test of size α has greater power.

Theorem (Neyman-Pearson Lemma)

$X_1 \cdots X_n \sim f(x|\theta)$ i.i.d.

Consider testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$, where the pdf or pmf corresponding to θ_i is $f(x|\theta_i)$, $i = 0, 1$. Then the most powerful test is

$L(\theta|\vec{x}) = \prod_{i=1}^{n} f(x_i|\theta)$

$\exists$ const C.

$$\text{to reject } H_0 \text{ if } \Lambda(\mathbf{x}) < c,$$

where $\Lambda(\mathbf{x}) = f(\mathbf{x}|\theta_0)/f(\mathbf{x}|\theta_1)$ is the likelihood ratio and constant $c > 0$. Further more the size of the test is given by

$= P(\text{Type I error}) = P(\text{reject } H_0 | H_0 \text{ true}) = P(\Lambda(\vec{X}) < c \,|\, \theta = \theta_0)$.

$X_i \sim f(x|\theta_0)$

$$\alpha = \int_R f(x|\theta_0)dx, \text{ where } R = \{x : \Lambda(x) < c\}.$$

to identify const C. Rejection region.

$H_0: \mu = \mu_0$ vs. $H_1: \mu > \mu_0$

Example 4.6. Let $X_1, \cdots, X_n \sim N(\mu, \sigma^2)$ i.i.d., σ^2 is known. Consider simple hypotheses $H_0: \mu = \mu_0$ vs $H_1: \mu = \mu_1$, where μ_0, μ_1 are given constants. The likelihood ratio test statistic

$\theta_0 = \mu_0$ $\quad L(\mu_0) = \prod_{i=1}^{n} f(x_i|\mu_0) = (2\pi\sigma^2)^{-\frac{n}{2}} e^{-\frac{\sum(x_i-\mu_0)^2}{2\sigma^2}}$

Test stat

$\bar{X} \sim N(\mu, \sigma^2/n)$

$$\Lambda(\mathbf{X}) = \frac{f(\mathbf{X}|\theta_0)}{f(\mathbf{X}|\theta_1)} = \frac{\exp[-\sum_{i=1}^{n}(X_i-\mu_0)^2/(2\sigma^2)]}{\exp[-\sum_{i=1}^{n}(X_i-\mu_1)^2/(2\sigma^2)]}$$

$L(\mu_1)$

$$= \exp[\frac{1}{2\sigma^2}\{2n\bar{X}(\mu_0-\mu_1)+n\mu_1^2-n\mu_0^2\}]$$

$= \exp\left(\frac{1}{2\sigma^2} 2n(\mu_0-\mu_1)\bar{X}\right) e^{n\mu_1^2 - n\mu_0^2}$.

$\mu_0 - \mu_1 < 0$

Assume that $\mu_0 < \mu_1$. $\Lambda(X)$ is small when $\bar{X}$ is large. By N-P Lemma, the MP test rejects H_0 when $\bar{X} > c$, c satisfies

$\Lambda(\bar{X})$ is montone decreasing function of $\bar{X}$.

$\alpha = P(\bar{X} > c|H_0) = P(\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} > \frac{c-\mu_0}{\sigma/\sqrt{n}}|H_0) \Rightarrow \frac{c-\mu_0}{\sigma/\sqrt{n}} = z(\alpha)$.

So the MP test of size α is to reject H_0 when $\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} > z(\alpha)$.

reject H_0 if $\quad c = \mu_0 + z(\alpha)\cdot\sigma/\sqrt{n}$ free of μ_1

$\Lambda(x) < c$, $\iff$ $\bar{X} > c^*$. $\quad \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} > \frac{c^*-\mu_0}{\sigma/\sqrt{n}}$ under H_0

$\sim N(0,1)$

$H_1: \theta = \theta_1 > \theta_0. \Rightarrow H_1: \theta > \theta_0$

Extension of Neyman-Pearson Lemma

In practice, a simple null hypothesis against a simple alternative is rarely encountered.

- Often consider the problem of testing $H_0: \theta = \theta_0$ versus $H_1: \theta > \theta_0$ based on a random sample from $f(x|\theta)$, where H_1 is a composite hypothesis.
- Uniformly most powerful (UMP) test. If a test of $H_0: \theta = \theta_0$ versus $H_1: \theta = \theta_1$ is a most powerful test for every $\theta_1 > \theta_0$ among all tests of size α, then the test is uniformly most powerful (UMP) for testing $H_0: \theta = \theta_0$ versus $H_1: \theta > \theta_0$.
- Approach to finding a UMP test:
  - Use the Neyman-Pearson Lemma to find a MP test of $H_0: \theta = \theta_0$ vs $H_1: \theta = \theta_1$ for some $\theta_1 > \theta_0$
  - If the form of the test is the same for all θ_1, then the test is UMP.

    Test stat, rejection rule, value of C.

Example 4.6 (revisit). Consider testing hypotheses

$$H_0 : \mu = \mu_0 \text{ vs } H_1 : \mu > \mu_0.$$

For any alternative value $\mu_1 > \mu_0$, we know that the MP test rejects H_0 when

$$\bar{X} > c \text{ and } c = \mu_0 + z(\alpha)\sigma/\sqrt{n},$$

which doesnt depend on μ_1. The rejection region is most powerful and the same for any alternative $\mu_1 > \mu_0$ Therefore, it is uniformly most powerful.

Note (One-sided Z-test): For one-sided hypothesis above, the UMP test of size α rejects H_0 when $\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} > z(\alpha)$.

$\Lambda(x) = \exp\left\{\frac{1}{2\sigma^2} 2n\underbrace{(\mu_0-\mu_1)}_{>0}\bar{X}\right\} \cdot \exp\{-2n\mu_0^2 + 2n\mu_1^2\}$

$H_0: \mu=\mu_0$, $H_1: \mu=\mu_1<\mu_0$

when $\mu_1<\mu_0 \Rightarrow (\mu_0-\mu_1)>0 \Rightarrow \Lambda(\vec{x})$ is monotone increasing function $\bar{X}$.

N-P Lemma, MP test is to reject H_0 when $\Lambda(\vec{x})<c$ $\Leftrightarrow \bar{X}<c$.

Example 4.7. In example 4.6,

1. consider testing $H_0 : \mu = \mu_0$ vs $H_1 : \mu < \mu_0$. What is the UMP rejection region?

   $\alpha = P(\text{Type I error}) = P(\bar{X}<c \mid H_0)$
   $= P\left(\underbrace{\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}}_{\sim N(0,1)} < \frac{c-\mu_0}{\sigma/\sqrt{n}} \mid H_0\right)$

   $\bar{X} \sim N(\mu_0, \sigma^2/n)$ under H_0.

   $$R = \{x : \bar{X} < c\}$$

   $c = \mu_0 - Z(\alpha)\cdot\sigma/\sqrt{n}$

   $\frac{c-\mu_0}{\sigma/\sqrt{n}} = -Z(\alpha)$

2. Consider testing $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$ The UMP rejection regions are different for $H_1 : \mu > \mu_0$ and $H_1 : \mu < \mu_0$.

	$H_1: \mu<\mu_0$	$H_1: \mu>\mu_0$
Test Stat:	$\bar{X}$	$\bar{X}$ ✓
Reject rule:	$\bar{X}<c$	$\bar{X}>c$.
	$c=\mu_0-Z(\alpha)\sigma/\sqrt{n}$	$c=\mu_0+Z(\alpha)\sigma/\sqrt{n}$,

   $\Rightarrow$ No UMP test.

   $$R = \left\{x : \left|\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}\right| < c\right\}$$

3. Consider the same hypotheses, but unknown σ^2. What are the corresponding UMP tests?

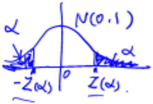
   S^2

   t-tests for μ: $T = \frac{\bar{X}-\mu_0}{S/\sqrt{n}}$,

   $H_0: \mu=\mu_0$ vs $H_1: \mu=\mu_1$

   $X\sim N(\mu_0, \sigma^2)$ not simple hypothesis.

   N-P Lemma ✗



pmf $f(x|p) = p^{x}(1-p)^{1-x}$.

Example 4.8. $X_1, \cdots, X_n \sim Bern(p)$ i.i.d., Consider simple hypotheses $H_0 : p = p_0$ vs $H_1 : p = p_1$, where $p_0 < p_1$ are given constants. The likelihood ratio is

if $p_0 > p_1$ $\Lambda(x)$ increasing function of $\sum X_i$; reject H_0 if $\sum X_i < c$

$$\Lambda(\mathbf{X}) = \frac{\overbrace{\prod_{i=1}^{n} p_0^{X_i}(1-p_0)^{1-X_i}}^{H_0}}{\underbrace{\prod_{i=1}^{n} p_1^{X_i}(1-p_1)^{1-X_i}}_{H_1}} = \left(\underbrace{\overbrace{\frac{p_0}{p_1}}^{<1}\overbrace{\frac{(1-p_1)}{(1-p_0)}}^{<1}}_{<1}\right)^{\sum_{i=1}^{n} X_i} \left(\frac{1-p_0}{1-p_1}\right)^{n}.$$

$\Lambda(x)$ is monotone decreasing funct of $\sum X_i$.

By N-P Lemma, MP test rejection region

$$R = \{\mathbf{x} : \Lambda(\mathbf{x}) < c_0\} = \{\mathbf{x} : \overbrace{\sum x_i}^{Y} > c\}.$$

$c = ?$

$\alpha \le P(\sum X_i > c \mid \sum X_i \sim Bino(n, p_0))$

Given α, we choose c (a positive integer) such that $P(X \in R|H_0) \le \alpha$.
Equivalently, $P(Y \overset{\le}{>} c | Y = \sum_i X_i \sim Bino(n, p_0))$.
Suppose $n = 20$, $p_0 = 0.25$. Then $P(Y > 9|Y \sim Bino(25, 0.25)) = 0.07$, $P(Y > 10|Y \sim Bino(25, 0.25)) = 0.03$. Therefore, $c = 10$. We reject H_0 when $Y > 10$ at level $\alpha = 0.05$.

$\alpha < P(Y > c \mid Y \sim Bino(n, p_0))$
$c = \min(y : P(Y > y \mid Y \sim Bino(n, p_0)))$

Note: If $y(obs) = 11 > c \Rightarrow$ reject H_0.
Equivalently, $\underbrace{P(Y > y(obs)|H_0)}_{p\text{-value}} = 0.01 < \alpha$.

$Y \sim Bino(n, p_0)$

$= P(Y < y(obs) | H_0) < \alpha$,
p-value.

p-value

- Let T be a test statistic and t_{obs} is the realized value of T. p-value is a measure of consistency between the data and H_0.

$$\text{p-value} = P(T \text{ is as or more extreme than } t_{obs}|H_0)$$

  A small p-value means the data are not consistent with H_0, and so is regarded as evidence against H_0.

- How can we make a decision based on the p-value?
  For given α, if p-value $< \alpha$, we reject H_0.
  if p-value $> \alpha$, we do not reject H_0.

- In Example 4.4, we reject H_0 when $\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} > z(\alpha)$.
  $\Leftrightarrow$ reject H_0 when p-value = $P\left(Z > \frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}}\right) < \alpha$.

p-value

- For an Z-test, let z_{obs} be the observation test statistic and $\Phi(x)$ be the cdf of $N(0,1)$. The p-value for testing H_0 vs H_1 is
  1. $P(Z > z_{obs}) = 1 - \Phi(z_{obs})$ if $H_1 : \mu > \mu_0$ Zobs > c. reject Ho,
  2. $P(Z < z_{obs}) = \Phi(z_{obs})$ if $H_1 : \mu < \mu_0$ Zobs < c.
  3. $1 - P(-|z_{obs}| < Z < |z_{obs}|) = 2[1 - \Phi(|z_{obs}|)]$ if $H_1 : \mu \neq \mu_0$.
- For an t-test, change Z to T, z_{obs} to t_{obs} and $N(0,1)$ distribution to t_{n-1} distribution.

Generalized likelihood ratio test

- Let $X_1, \cdots, X_n$ be i.i.d. with pdf $f(x|\theta)$ for $\theta \in \Omega$., Consider two-sided hypotheses

$$H_0 : \underbrace{\theta = \theta_0}_{\text{simple}} \text{ vs } H_1 : \underbrace{\theta \neq \theta_0}_{\text{composite.}},$$

where θ_0 is a specified value.

Recall the likelihood $L(\theta) = \prod_{i=1}^n f(X_i|\theta)$. Let $\hat\theta$ be the MLE of θ. Consider the ratio of two likelihood functions

$$\Lambda = \frac{\overbrace{L(\theta_0)}^{\text{small}}}{\underbrace{L(\hat\theta)}_{\max_\theta L(\theta)}} \leq 1$$

$L(\theta_0|\vec{x})$ under H_0

under $H_0 \cup H_1$

Rejection rule: small value of Λ is in favor of H_1. Thus

Reject H_0 if $\Lambda < c$, where c s.t. $\alpha = P(\Lambda < c|H_0)$.

$X_1 \cdots X_n \sim N(\mu, \sigma^2)$ unknown. $\theta = (\mu, \sigma^2)$

$H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$
composit

$\max_{\sigma^2} L(\theta|\vec{X})$ under $H_0: (\mu_0, \sigma^2)$ $= L(\mu_0, \hat{\sigma}_0^2)$

$\hat{\sigma}_0^2$ is MLE of σ^2, $N(\mu_0, \sigma^2)$, $\hat{\sigma}_0^2 = \frac{1}{n}\sum(X_i - \mu_0)^2$

$L(\hat{\mu}, \hat{\sigma}^2 | \vec{x})$, where $\hat{\mu} = \bar{X}$, $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$

- In general, when testing a composite null versus a composite alternative, the likelihood ratio (LR) is defined by

$\Lambda = \frac{L(\mu_0, \hat{\sigma}_0^2)}{L(\hat{\mu}, \hat{\sigma}^2)}$

H_0 & H_1 Composit

$$\Lambda = \frac{max_{\theta \in \Theta_0} L(\theta)}{max_{\theta \in \Theta} L(\theta)}, \quad \leq 1.$$

Λ small reject H_0.
$\Lambda < C$.

where Θ_0 is the parameter space under H_0 and Θ is the whole parameter space of θ.

$\Lambda = \frac{L(\theta_0)}{L(\hat{\theta})}$, $\log\Lambda = \log L(\theta_0) - \log L(\hat{\theta})$

- Properties of Λ

Taylor expansion at $\hat{\theta}$: $\log L(\theta) = \log L(\hat{\theta}) + \frac{d}{d\theta}\log L(\hat{\theta})(\theta - \hat{\theta}) + \frac{1}{2!}\frac{d^2}{d\theta^2} l(\hat{\theta})(\theta - \hat{\theta})^2 + \cdots$

Let $\theta = \theta_0$. $\log L(\theta_0) = \log L(\hat{\theta}) + \frac{1}{2} l''(\hat{\theta})(\theta_0 - \hat{\theta})^2 + \cdots$

1. $\Lambda(\mathbf{X}) \in [0, 1]$
2. Small values of Λ are evidence against H_0
3. Under some smoothness conditions, the asymptotic null distribution of $-2\log(\Lambda)$ is $\chi^2_{r-r_0}$ distribution, where r is the number of free parameters in Θ_1, r_0 is the number of free parameters in Θ_0.

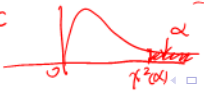
   $\hat{\theta} \overset{a}{\sim} N(\theta_0, \frac{1}{nI(\theta_0)})$

   $-l''(\hat{\theta})$, $r = 1$, $r_0 = 0$.
4. Decision rule: reject H_0 when $-2\log(\Lambda) > \chi^2_{r-r_0}(\alpha)$.

χ^2_{df} $df = r - r_0$.

reject H_0

$\Lambda < c \iff -2\log\Lambda > c$



Testing hypothesis concerning mean and variance in Normal distribution

Example 4.9. $X_1, \cdots, X_n \sim N(\mu, \sigma^2)$ i.i.d., Consider testing $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$ using the generalized LR method.

$\theta = \mu$, $\Theta_0 = \{\mu_0\}$ and $\Theta = \mathbb{R}$.

$$\Lambda = \frac{max_{\theta \in \Theta_0} L(\theta)}{max_{\theta \in \Theta} L(\theta)} = \frac{L(\mu_0)}{L(\hat{\mu})},$$

where $\hat{\mu} = \bar{X}$.

$$\Lambda < c \quad \Leftrightarrow \quad \left| \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \right| > c^*$$

Choose $c^* = z(\alpha/2)$ for a test with significance level α.

Example 4.10. Compare variances in two normal distributions $H_0 : \sigma_1^2 = \sigma_2^2 = \sigma^2$ vs $H_1 : \sigma_1^2 \neq \sigma_2^2$ using the generalized LR test. E.g.,

1. If two professors grading exams have the same variation in their grading.
2. A supermarket might be interested in the variability of check-out times for two cashers.

Two independent random samples of size n_1 and n_2 are collected from $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2$ respectively. Note that $\theta = (\mu_1, \mu_2, \sigma_1^2, \sigma_2)$, $\Theta_0 = \{-\infty < \mu_1, \mu_2 < \infty, \sigma^2 > 0\}$ and $\Theta = \{-\infty < \mu_1, \mu_2 < \infty, \sigma_1^2 > 0, \sigma_2^2 > 0\}$.

$$\Lambda = \frac{max_{\Theta_0} L(\theta)}{max_{\Theta} L(\theta)} = \frac{L(\hat{\mu}_1, \hat{\mu}_2, \hat{\sigma}^2)}{L(\hat{\mu}_1, \hat{\mu}_2, \hat{\sigma}_1^2, \hat{\sigma}_2^2)},$$

where $\hat{\mu}_i = \bar{X}_i$, $\hat{\sigma}_i^2 = S_i^{*2}$ for $i = 1, 2$, and $\hat{\sigma}^2 = \frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2}$.

$$\Lambda < c \quad \Leftrightarrow \quad F = S_1^2/S_2^2 < c_1 \text{ or } > c_2,$$

$F \sim \mathrm{F}_{n_1-1, n_2-1}$ under H_0. Therefore, we reject H_0 if

$$F_{obs} < F_{n_1-1,n_2-1}(1-\alpha/2) \text{ or } F_{obs} > F_{n_1-1,n_2-1}(\alpha/2).$$